

Simulating disease transmission

(Part 2)

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SCIENCE AND
EDUCATION
FOR
SUSTAINABLE
LIFE

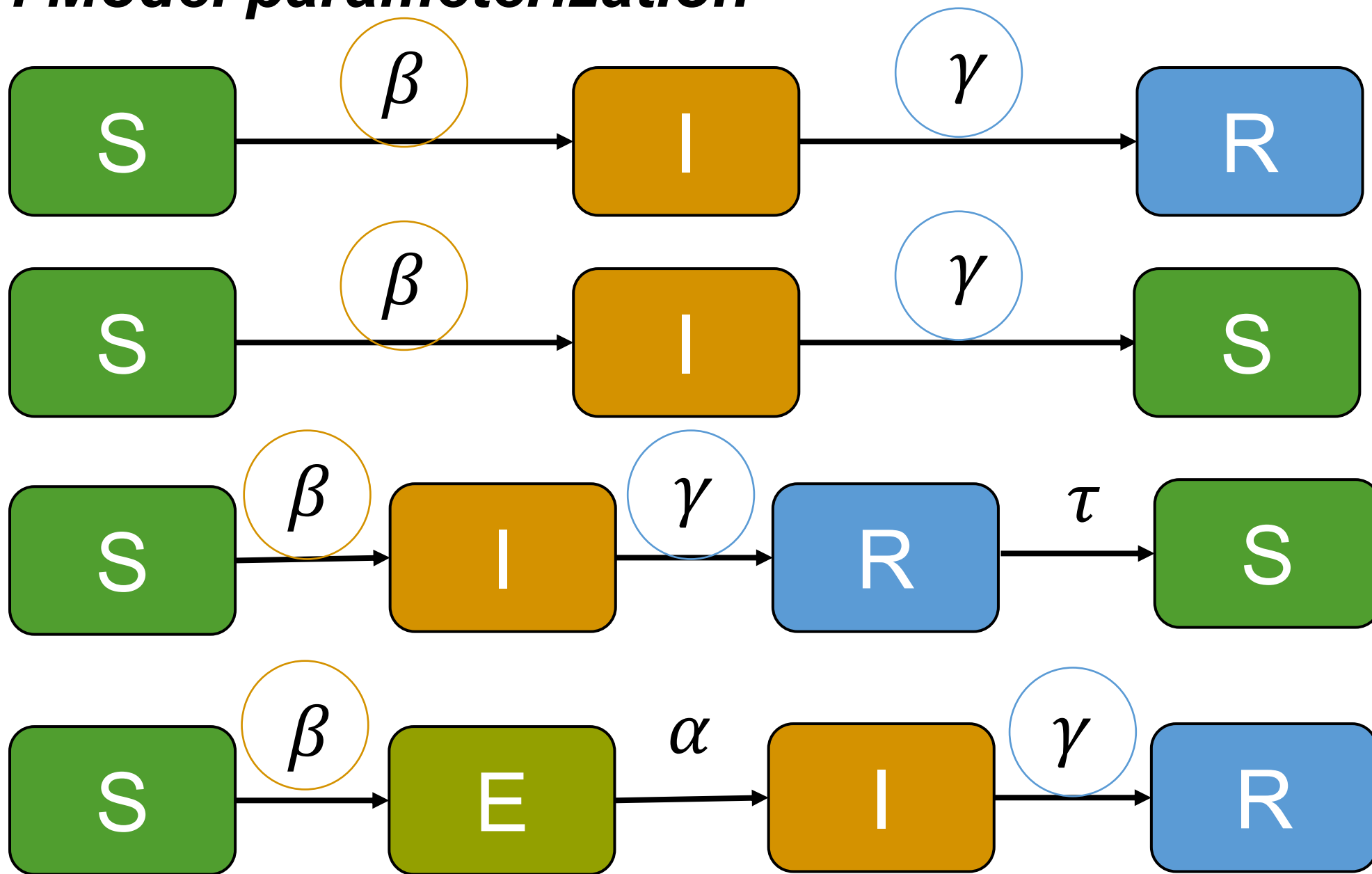
Overview

This lecture will be in 3 parts:

- 1. Model parameterization*
- 2. Introduction to individual-based models*
- 3. Introduction to spatial transmission models*

1. Model parameterization

1. Model parameterization



1. Model parameterization

- In deterministic settings:



- Parameters are generally chosen to minimize the deviation between observed and simulated epidemics.

- In stochastic settings:



- Parameters are generally chosen to maximize the likelihood of the model generating the observed data.



1. Model parameterization

- Estimating infection rate (β):

$$\log(\widehat{I_N}) = \log(\beta) + \log\left(\frac{S_{int} I_{int}}{N_{int}}\right)$$

Number of susceptible individuals

Number of infected individuals

Total number of individuals

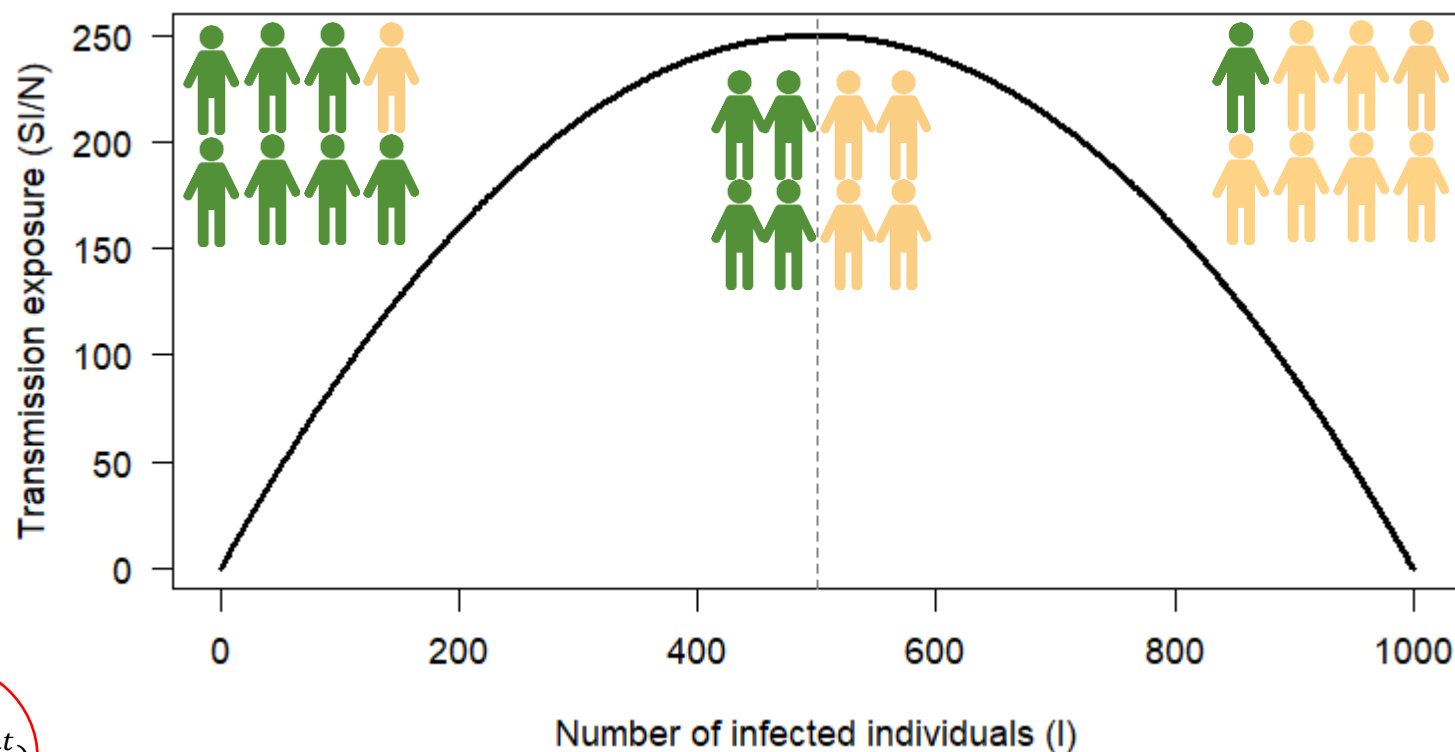
Number of new infections per sampling interval



Carsten Thure Kirkeby

1. Model parameterization

- Estimating infection rate (β):



$$\widehat{\log(I_N)} = \log(\beta) + \log\left(\frac{S_{int}I_{int}}{N_{int}}\right)$$

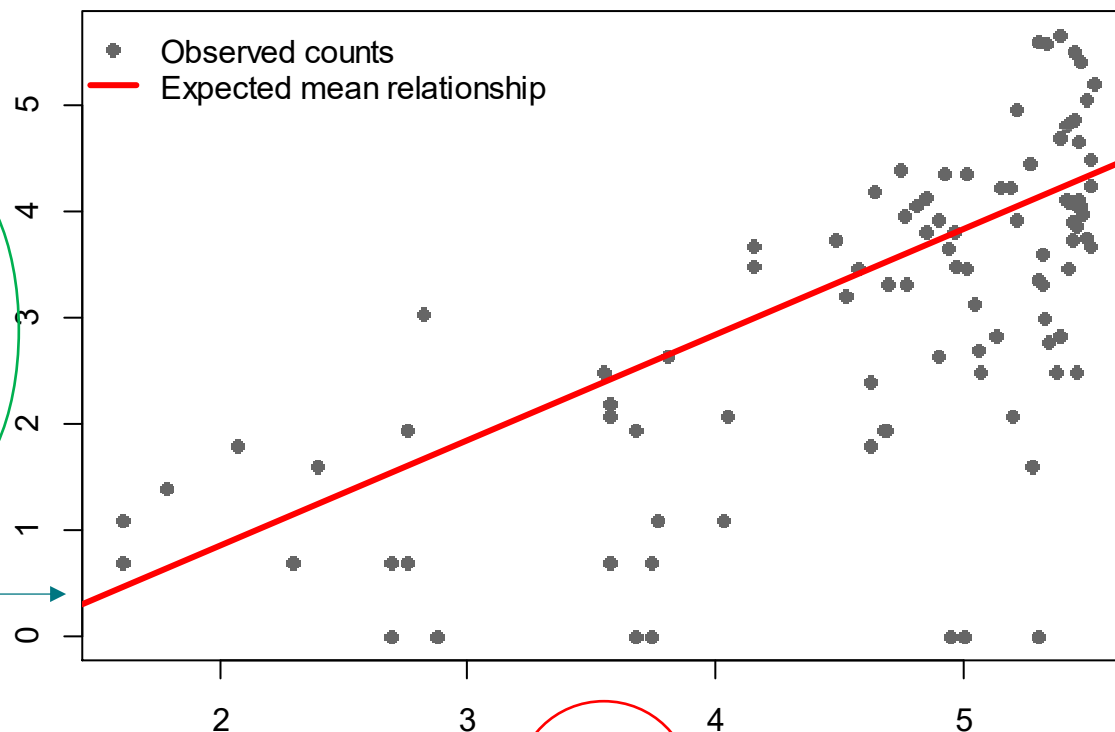
(Kirkeby et al., 2017)

1. Model parameterization

- Estimating infection rate (β):

$$\beta = 0.3$$

$$\hat{\beta} = 0.3133$$



$$\widehat{\log(I_N)} = \log(\beta) + \log\left(\frac{S_{int} I_{int}}{N_{int}}\right)$$

log(SI/N)

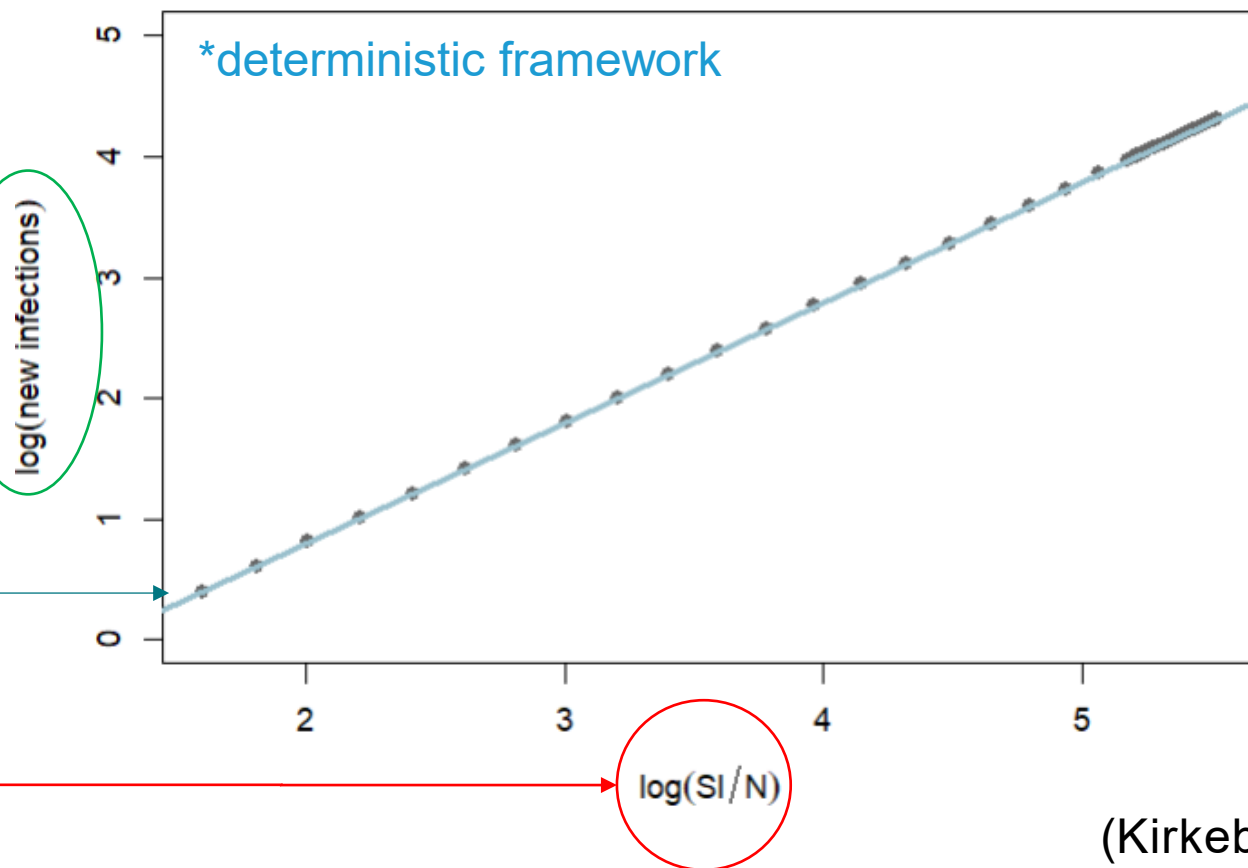
(Kirkeby et al., 2017)

1. Model parameterization

- Estimating infection rate (β):

$$\beta = 0.3$$

$$\hat{\beta} = 0.3$$



(Kirkeby et al., 2017)

1. Model parameterization

- Estimating recovery rate (γ):

$$\log(\widehat{R}_N) = \log(\gamma) + \log(I_{int})$$

Number of infected individuals

Number of new recoveries per sampling interval



Carsten Thure Kirkeby

(Kirkeby et al., 2017)

1. Model parameterization

- Estimating R_0 :

$$R_0 = \frac{\beta}{\gamma}$$

Infection rate

Recovery rate

- $R_0 > 1$:
 - Each infected individual generates, on average, more than one secondary infection
 - Transmission can increase.
- $R_0 < 1$:
 - Each infected individual generates, on average, fewer than one secondary infection.
 - Transmission is expected to decline.
- $R_0 = 1$:
 - Each infected individual replaces itself with approximately one new infection
 - Prevalence is expected to be constant.

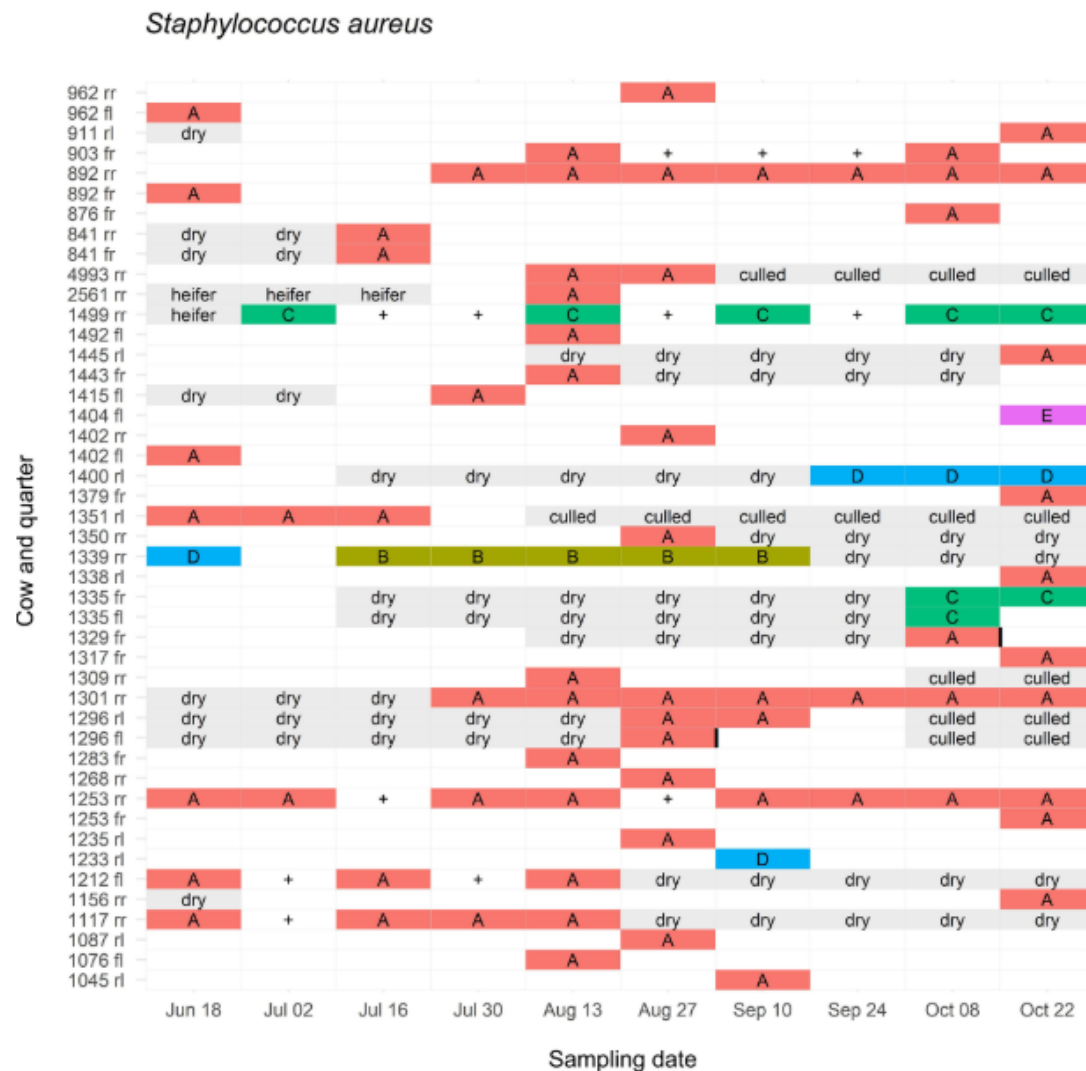
UNIVERSITY OF COPENHAGEN
DEPARTMENT OF VETERINARY AND ANIMAL SCIENCES

Transmission pathways and infection durations of bovine mastitis pathogens

PhD Thesis 2023

Svenja Woudstra

This thesis has been submitted to the Graduate School of Health and Medical Sciences, University of Copenhagen, 31st of May 2023



(Woudstra et al., 2017)

1. Model parameterization

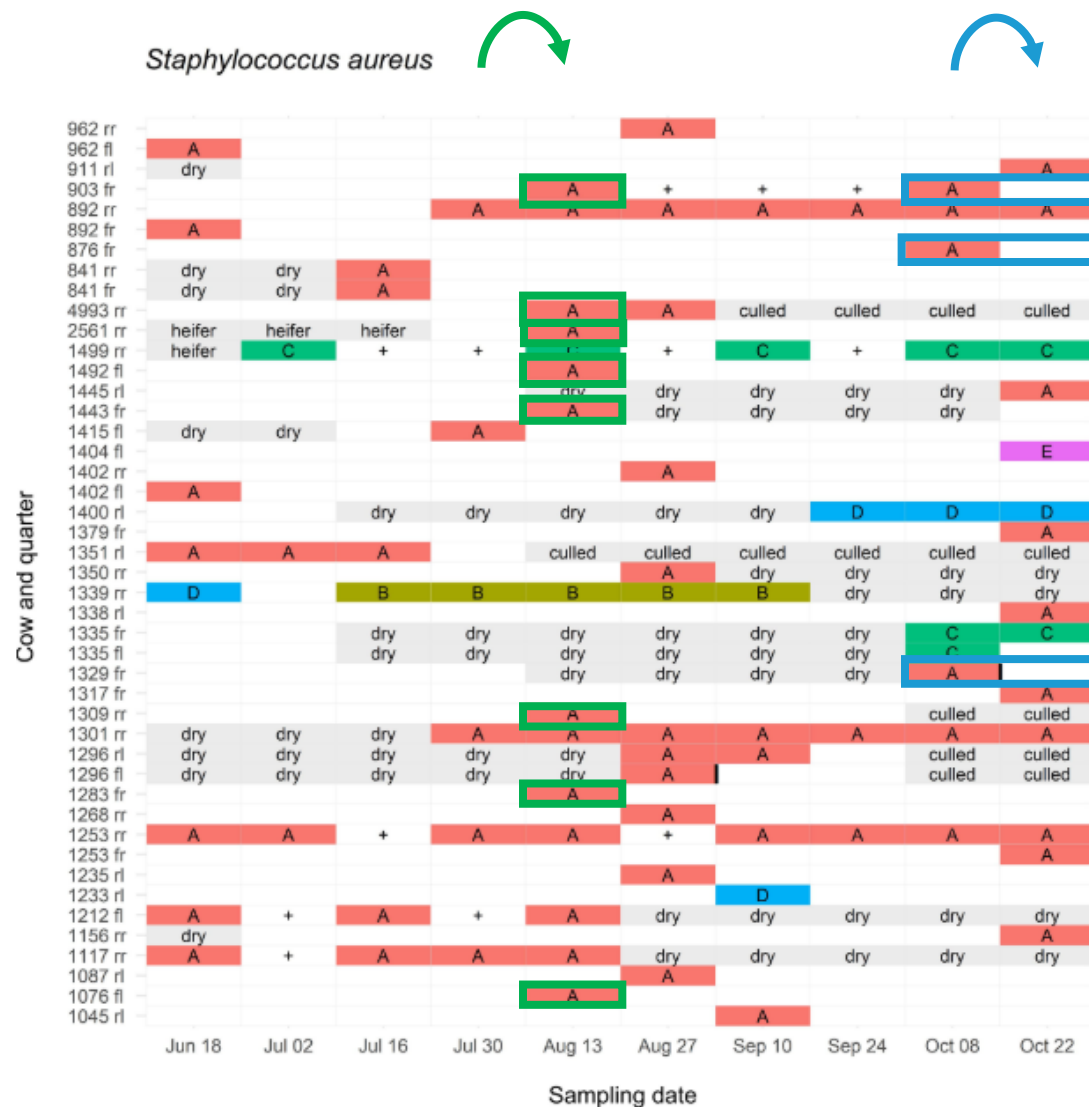
$$\log(\widehat{I_N}) = \log(\beta) + \log\left(\frac{S_{int}I_{int}}{N_{int}}\right)$$

$\beta: 0.75$

$$\log(\widehat{R_N}) = \log(\gamma) + \log(I_{int})$$

$\gamma: 0.58$

$$R_0 = \frac{\beta}{\gamma} = 1.29$$



(Woudstra et al., 2017)

2. Introduction to individual-based models

1. General introduction to Epidemic models

Compartmental models

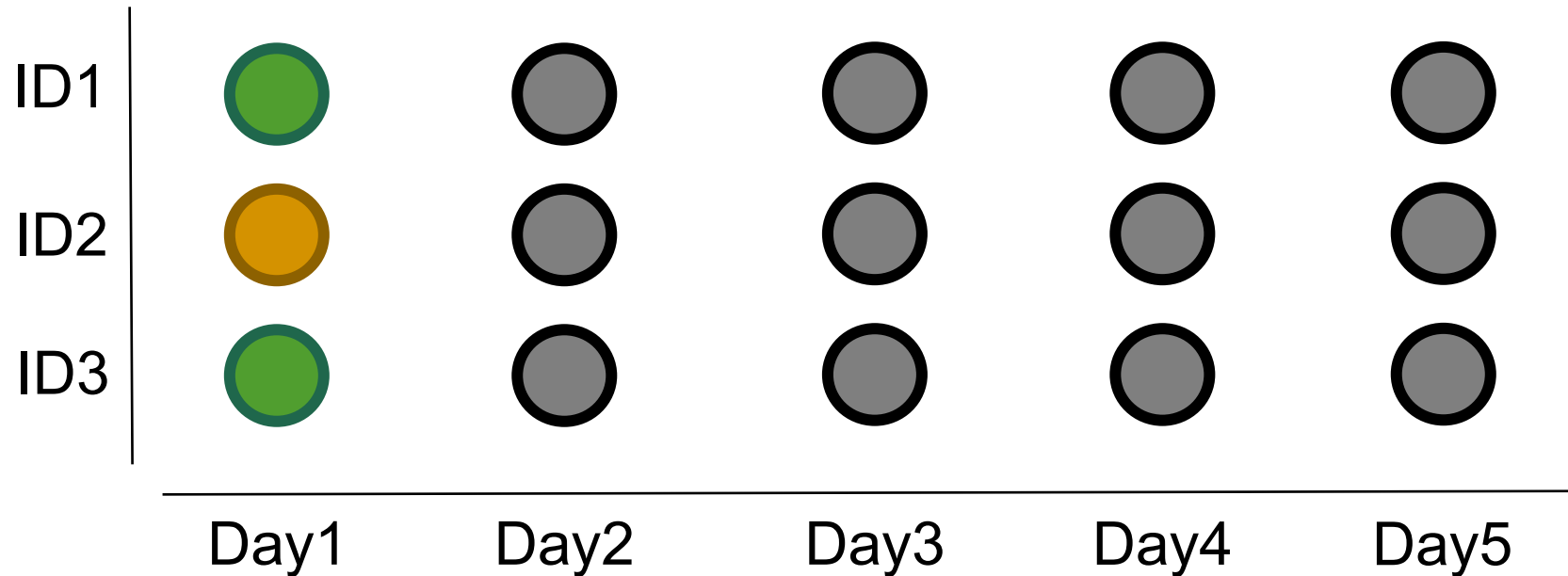
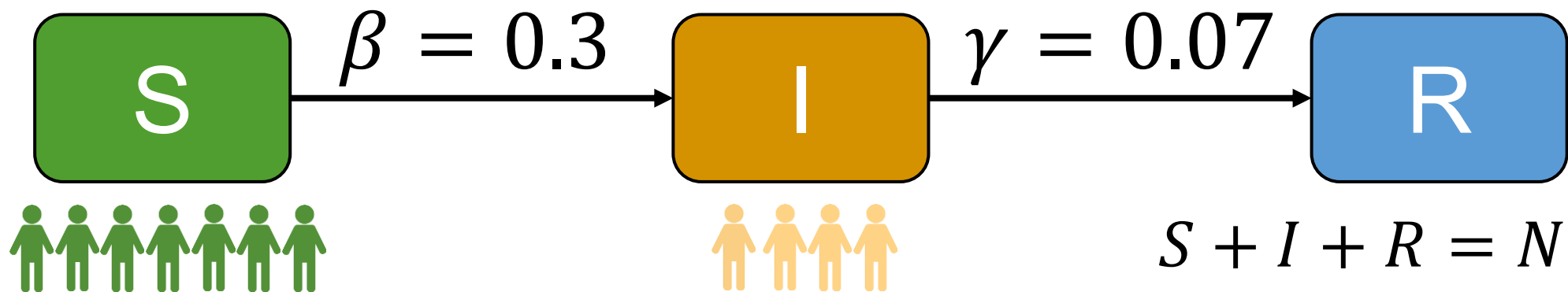


Individual-based models

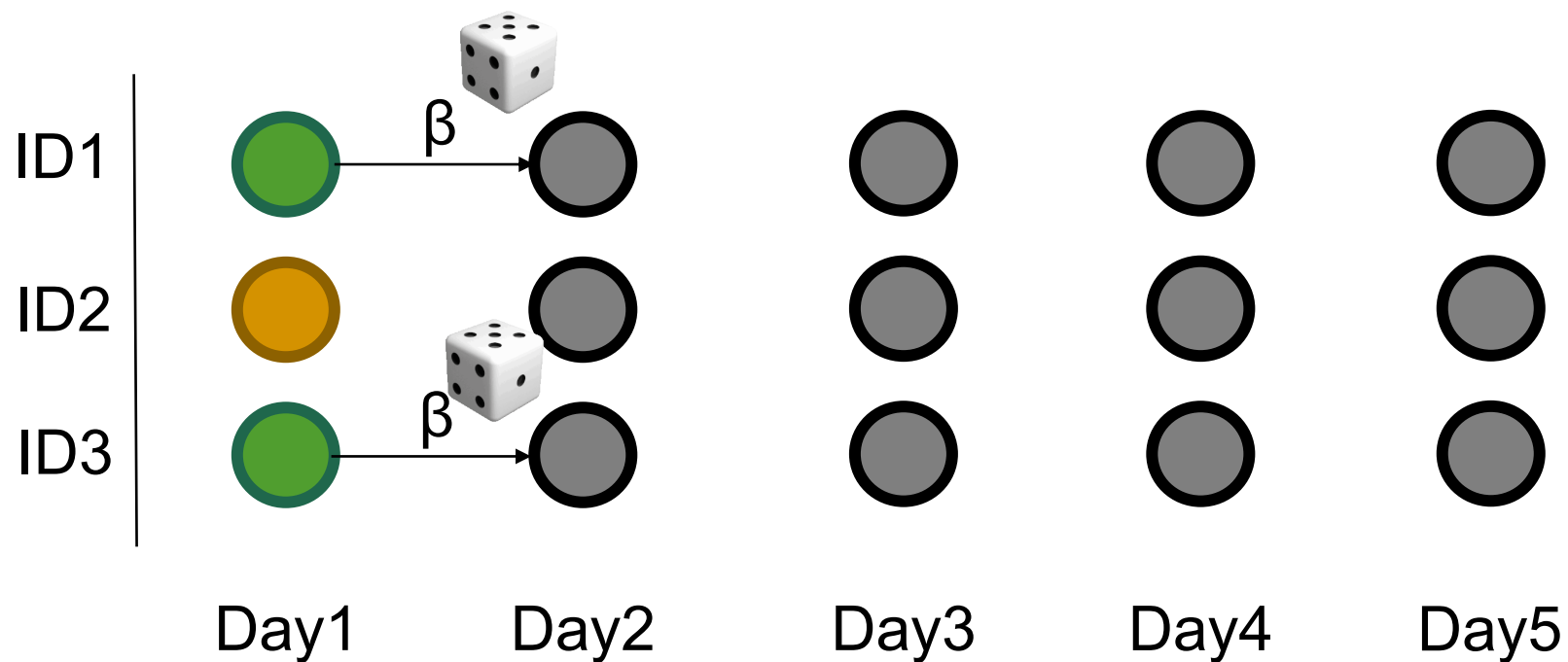
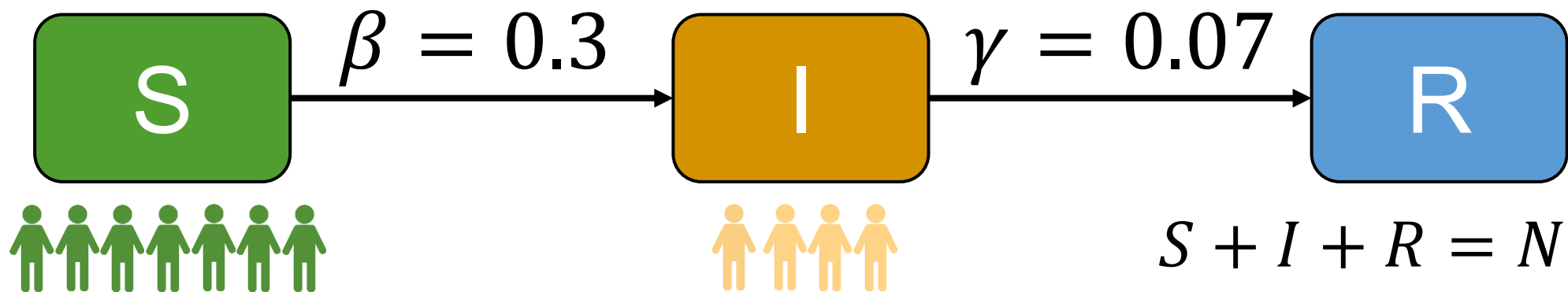


	Compartmental models	Individual-based models
Fundamental question	What happens to groups of individuals ?	What happens to each individual ?
Representation	Individuals grouped into compartments (e.g. S = 80 , I = 20)	Each individual represented explicitly (e.g. Cow 1 = S , Cow 2 = I)
Heterogeneity	Represented by adding groups/compartments	Can be assigned directly to each individual
Best suited to	Population-level dynamics and relatively homogeneous populations	Individual heterogeneity, specific contacts, spatial structure or complex behaviour
Computational cost	Low	High

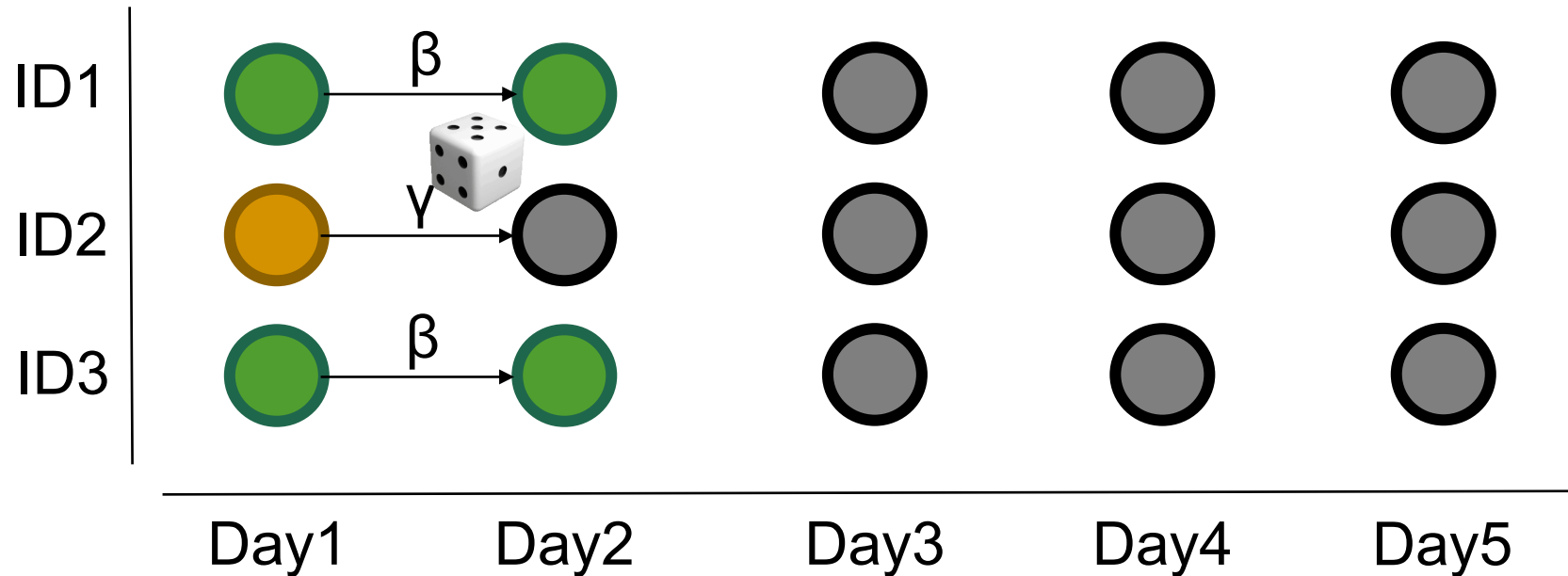
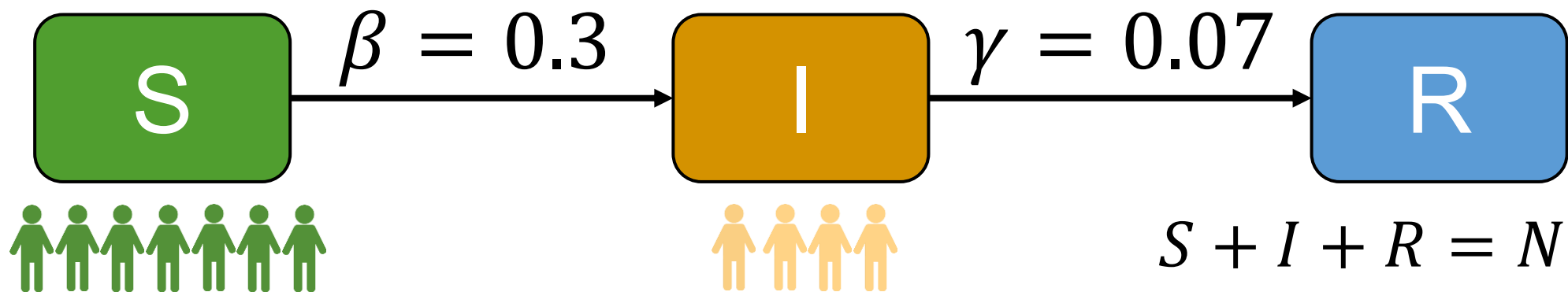
2. Introduction to individual-based models



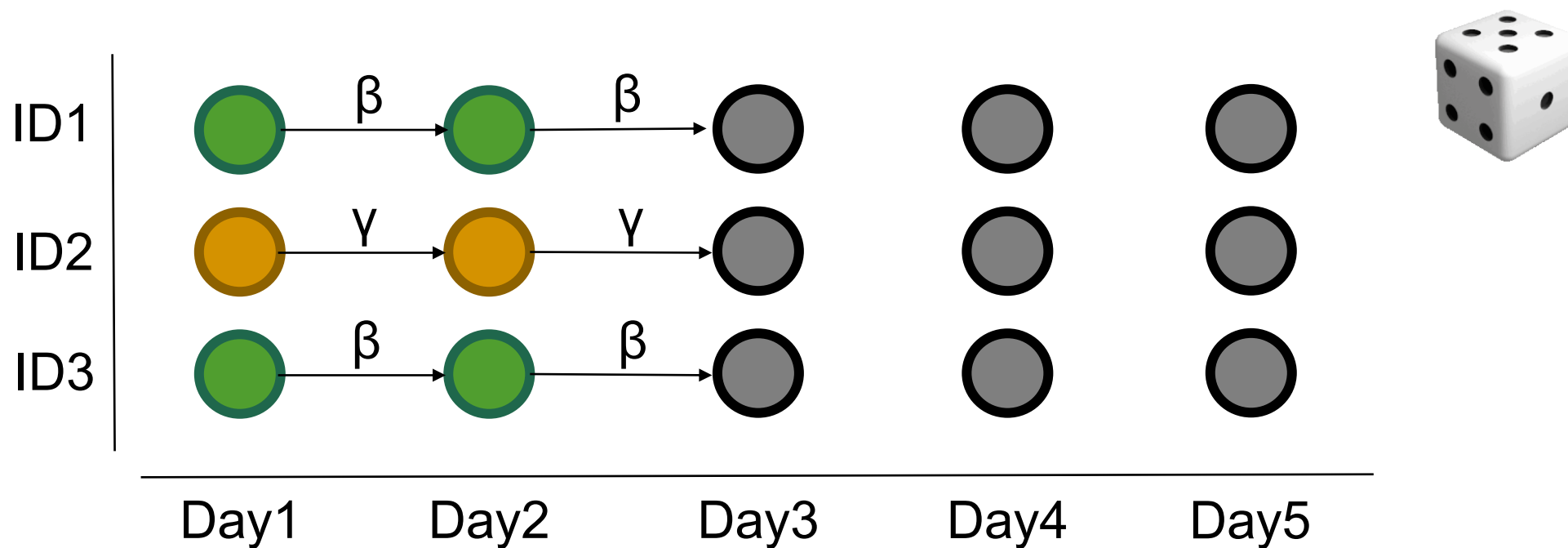
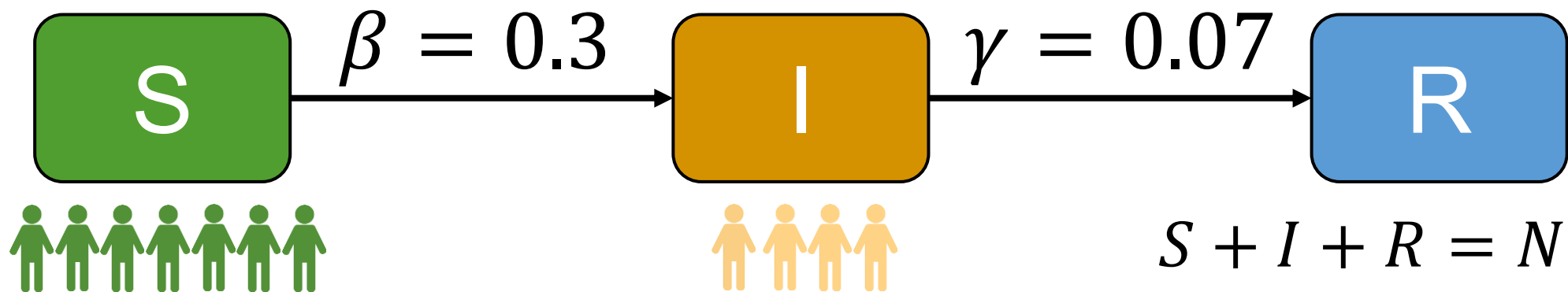
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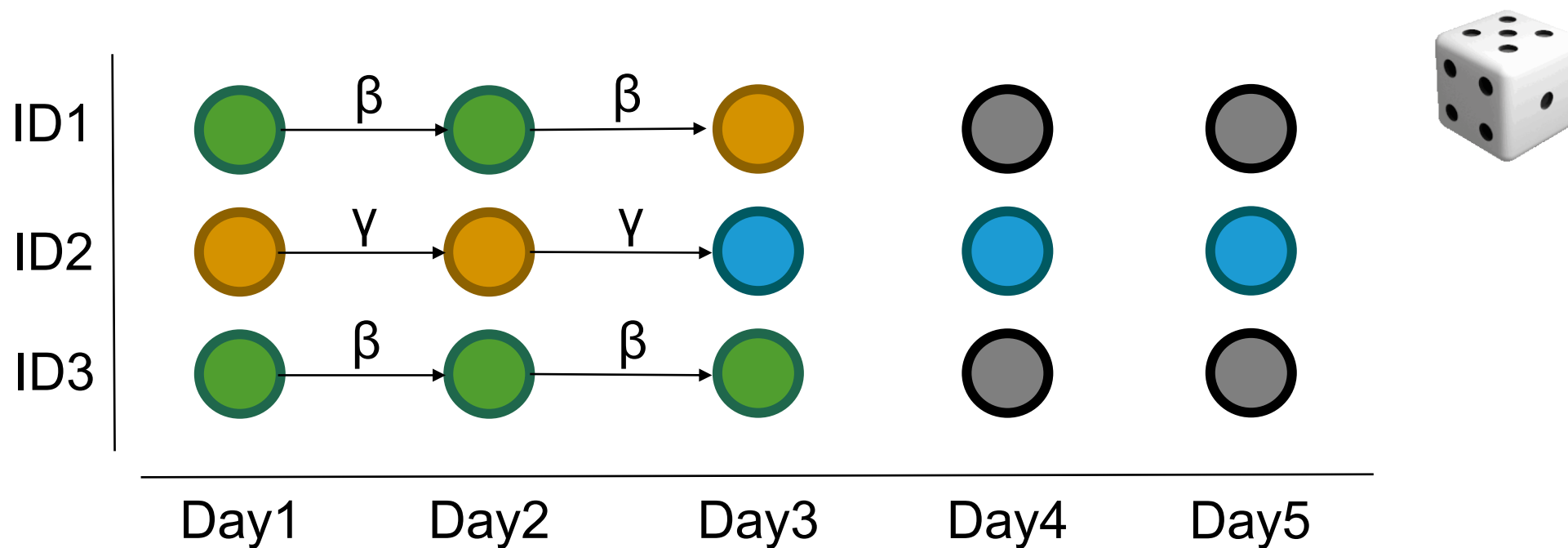
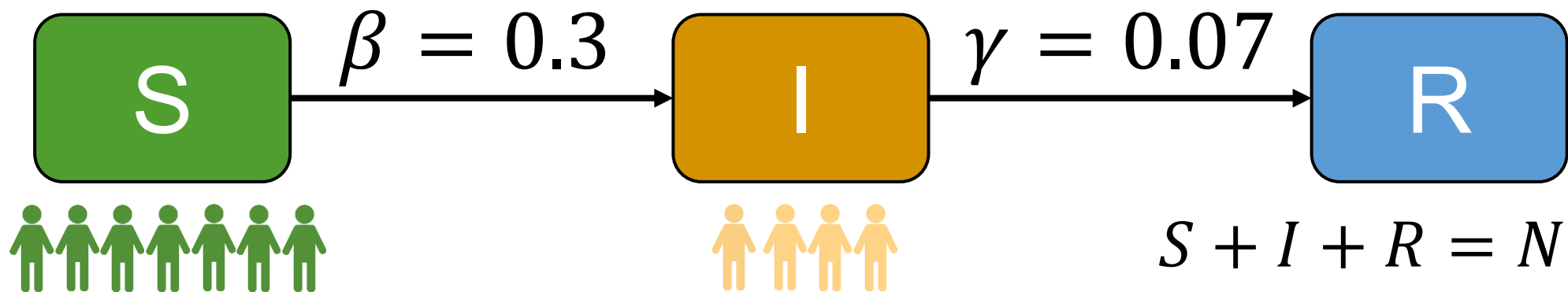
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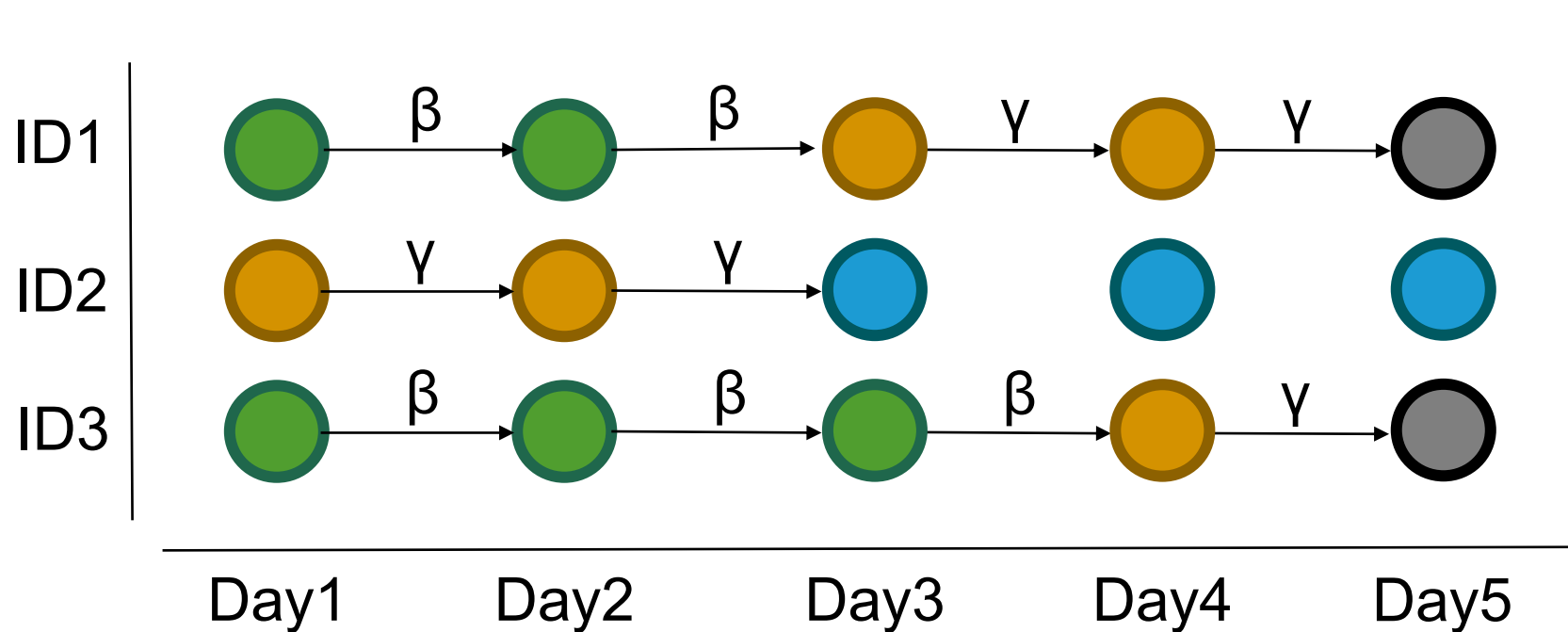
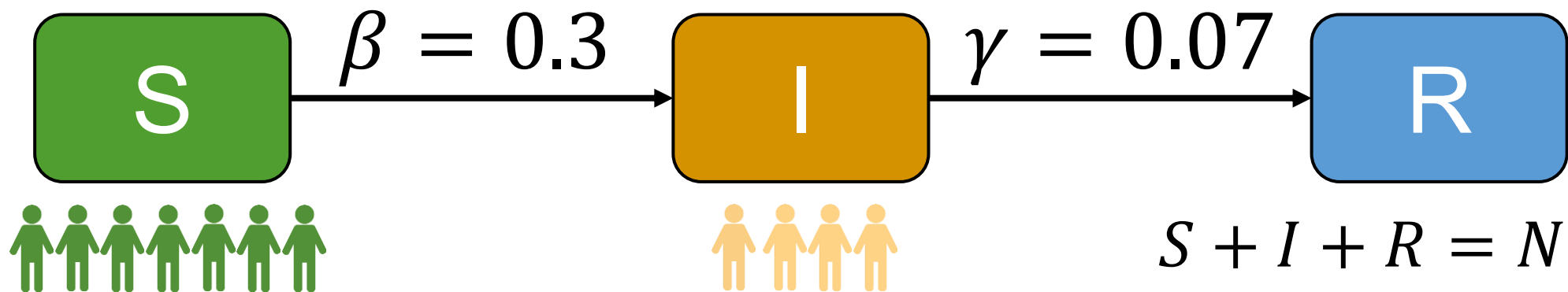
2. Introduction to individual-based models



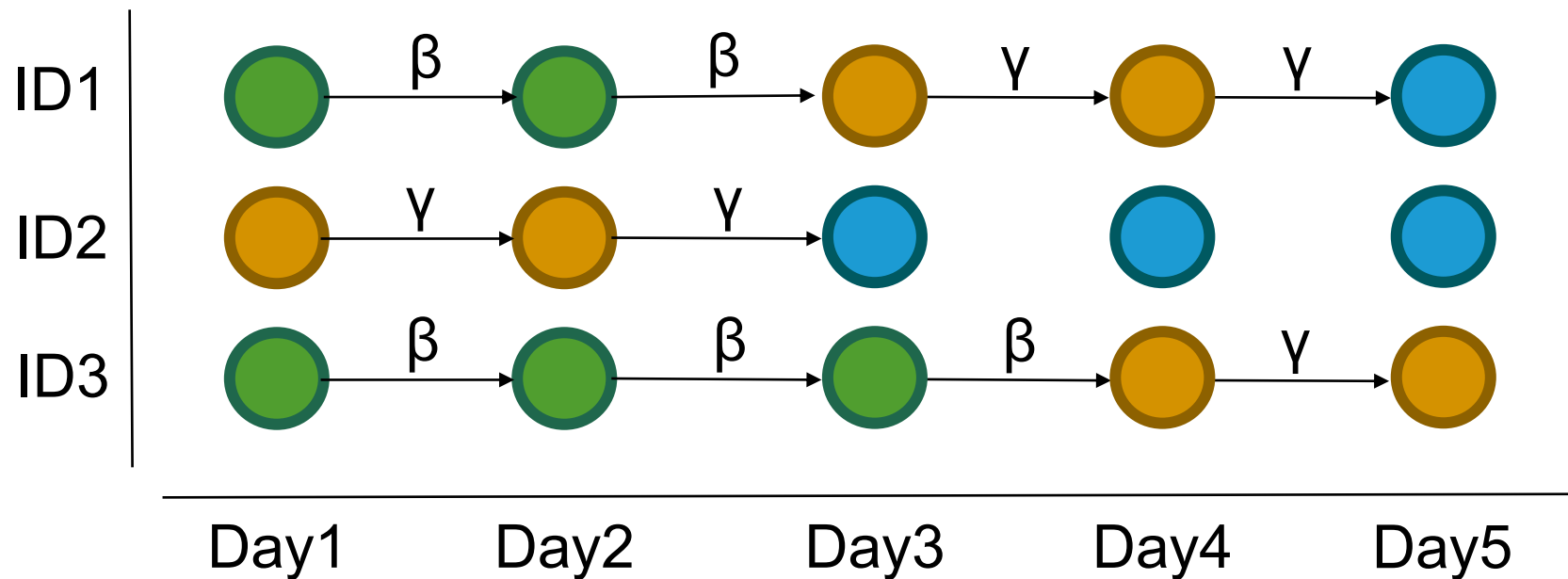
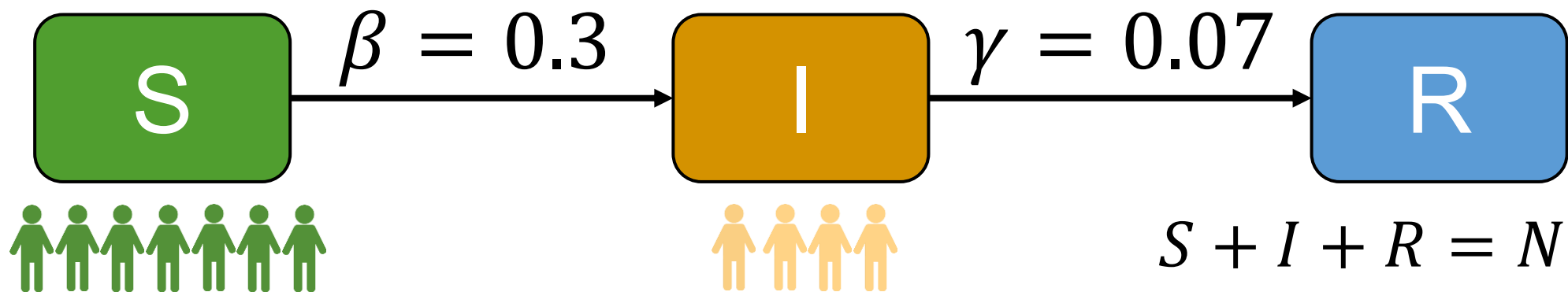
2. Introduction to individual-based models



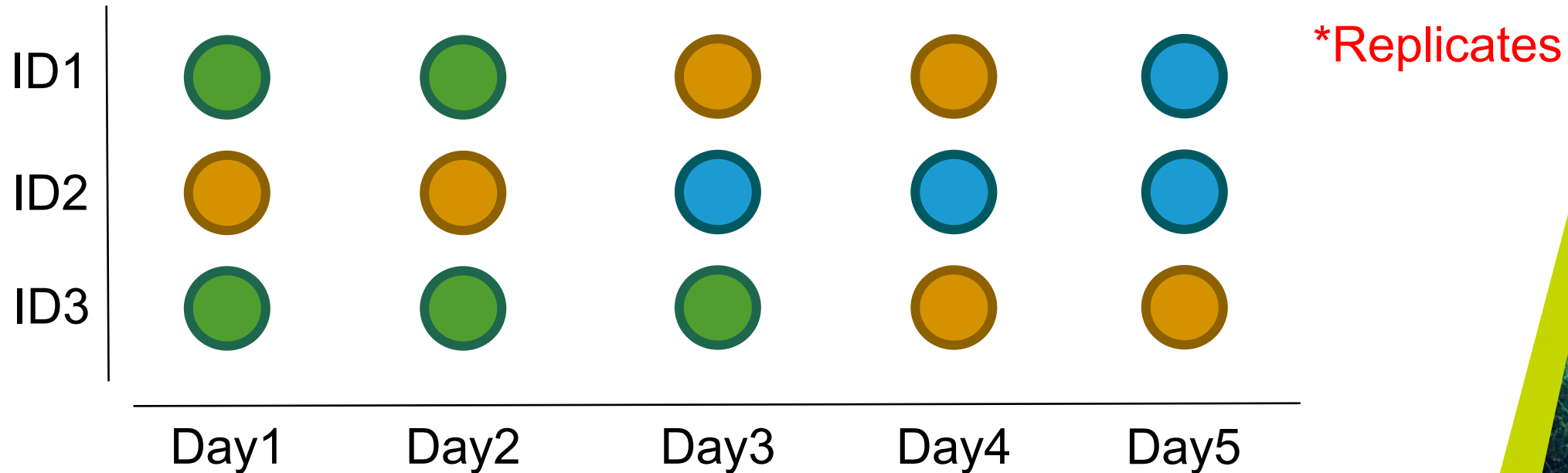
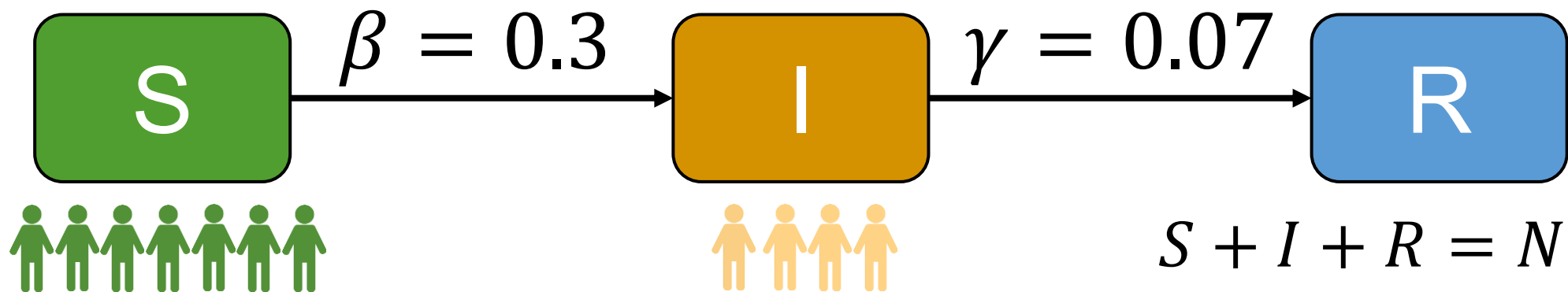
2. Introduction to individual-based models



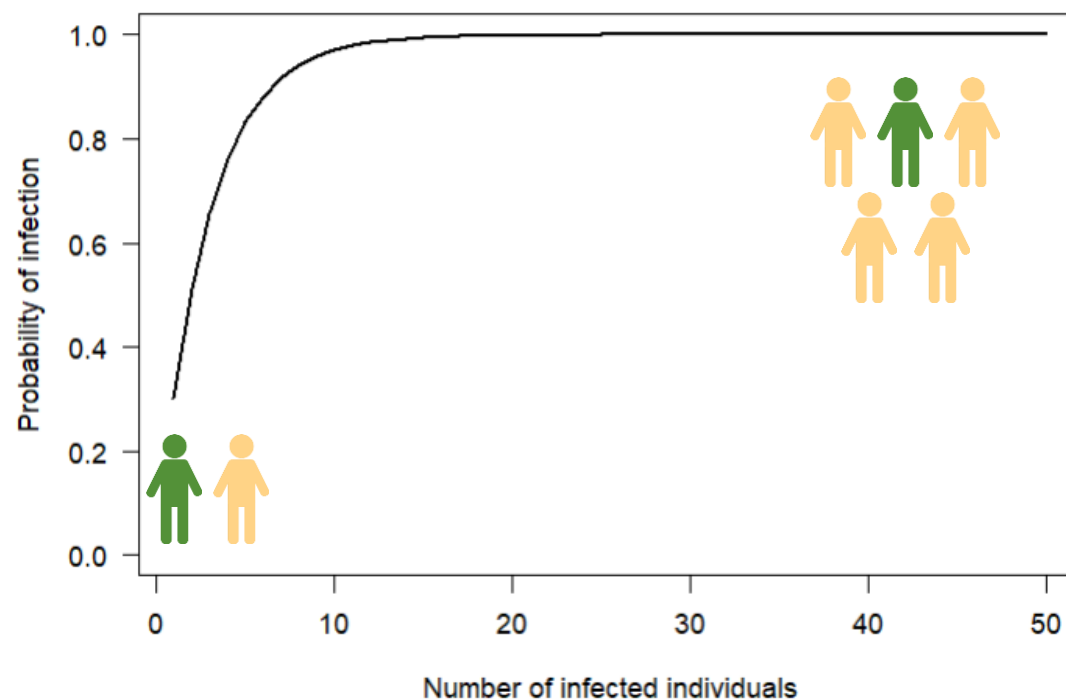
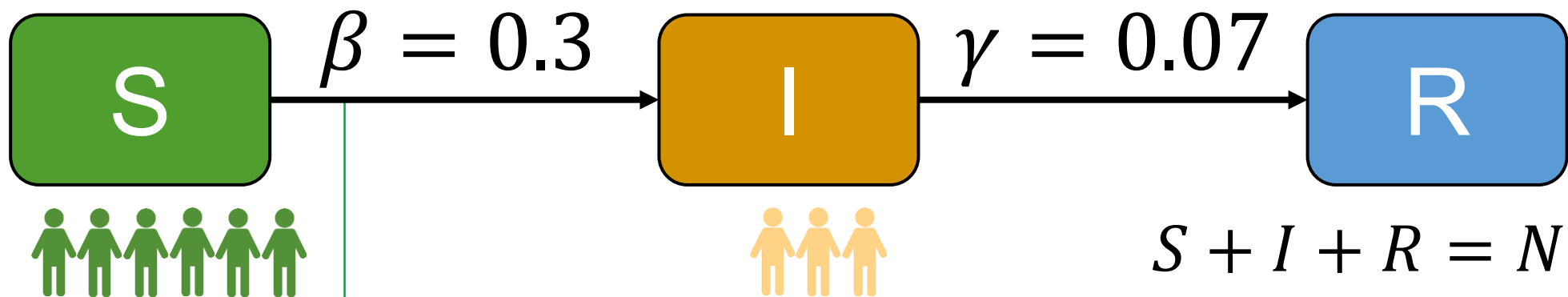
2. Introduction to individual-based models



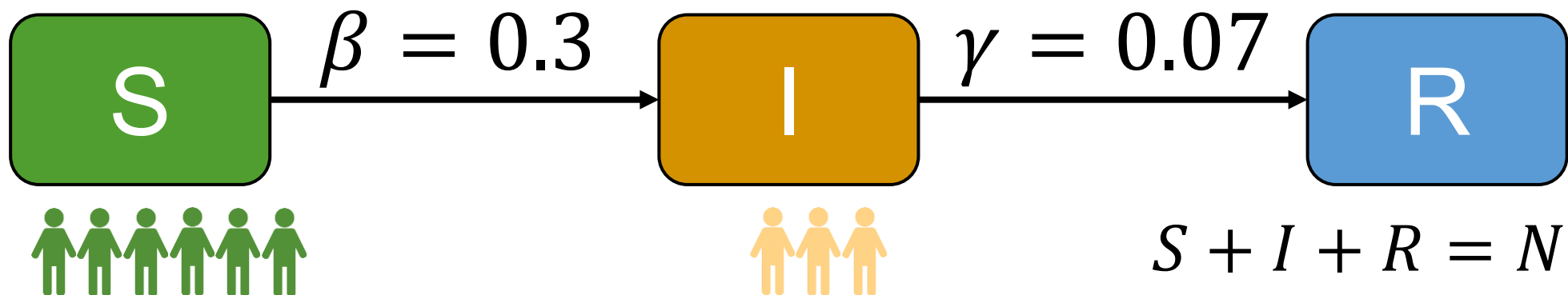
2. Introduction to individual-based models



2. Introduction to individual-based models



2. Introduction to individual-based models



β ← “Probability of infection”

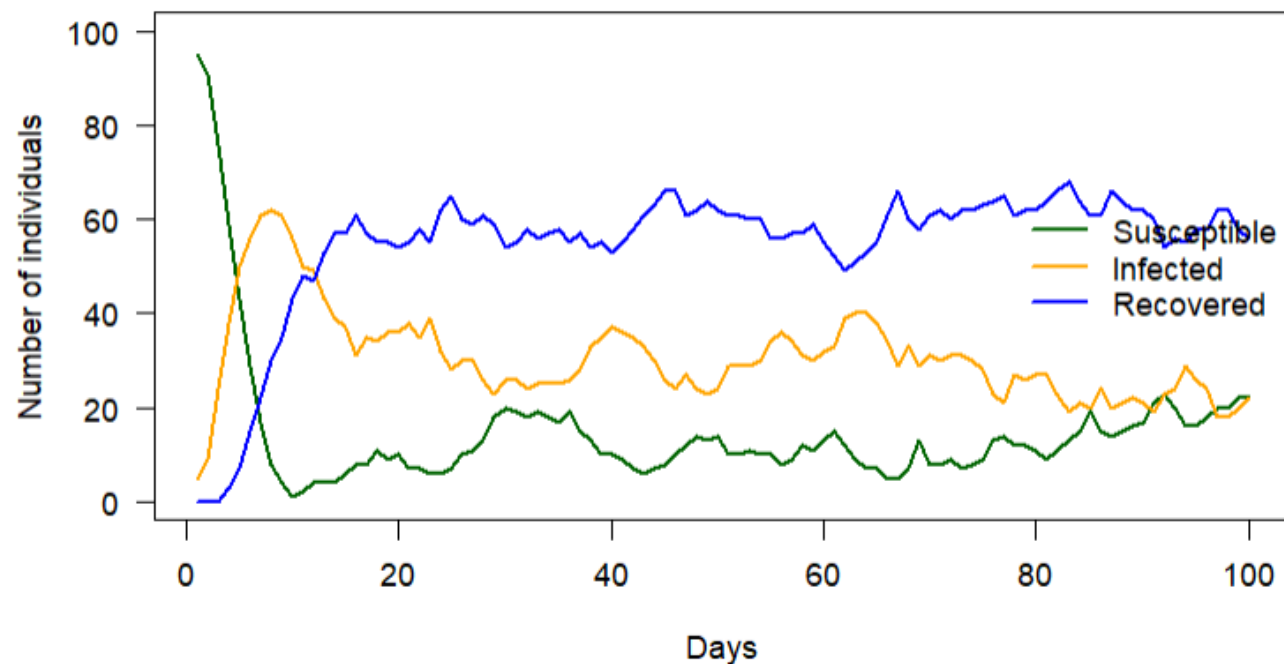
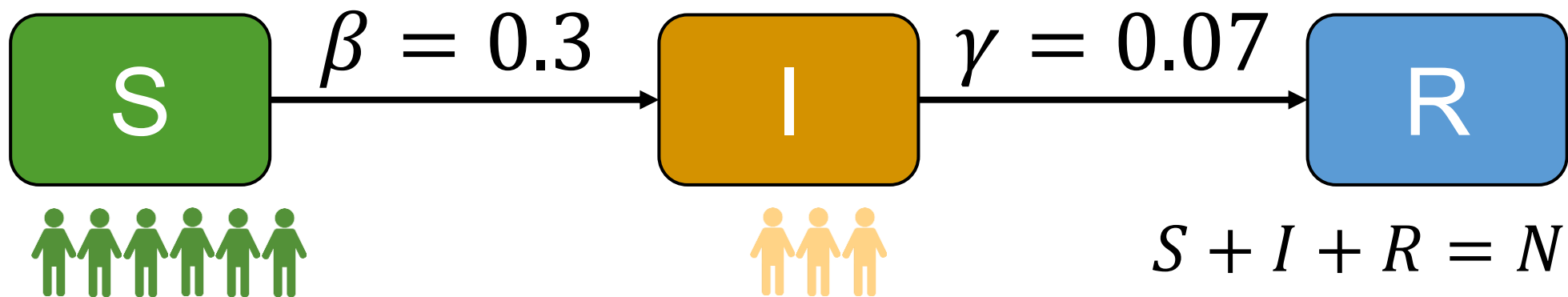
$(1 - \beta)$ ← Probability of not getting infected

$(1 - \beta)^k$ ← Probability of not getting infected by any of the infected individuals

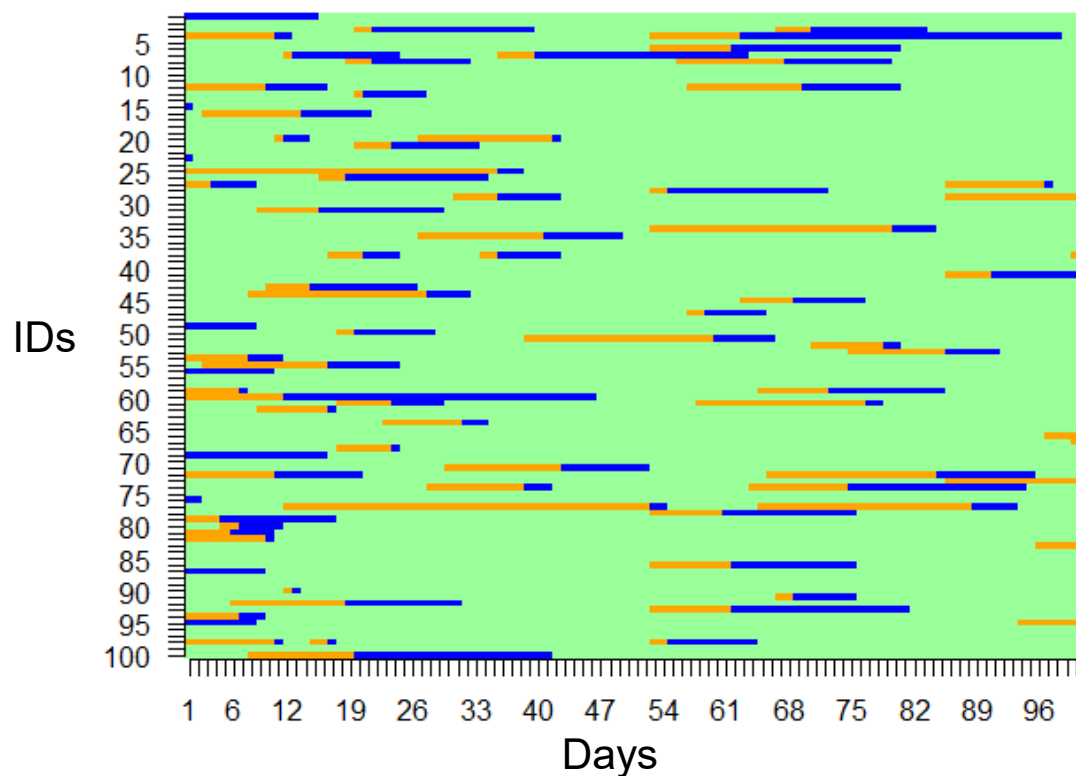
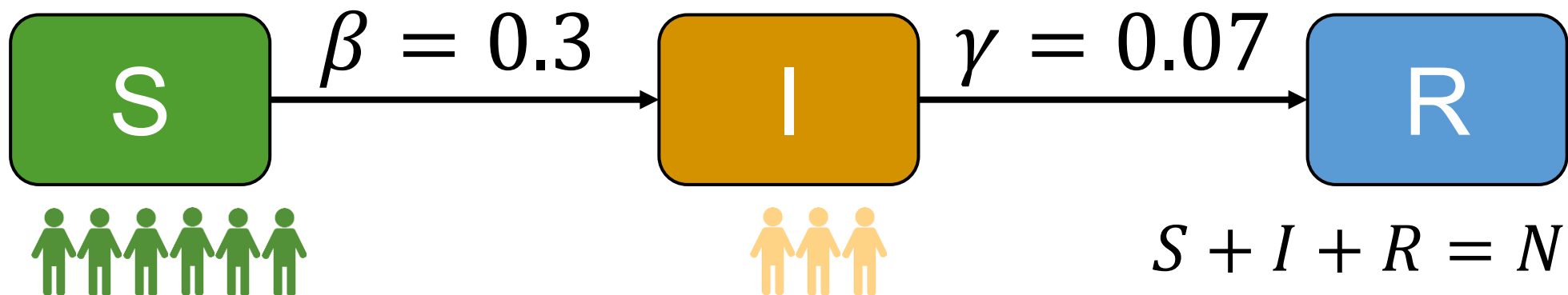
$$P(\text{infection}) = 1 - (1 - \beta)^k$$

↑ Probability of getting infected by at least one of the infected individuals

2. Introduction to individual-based models

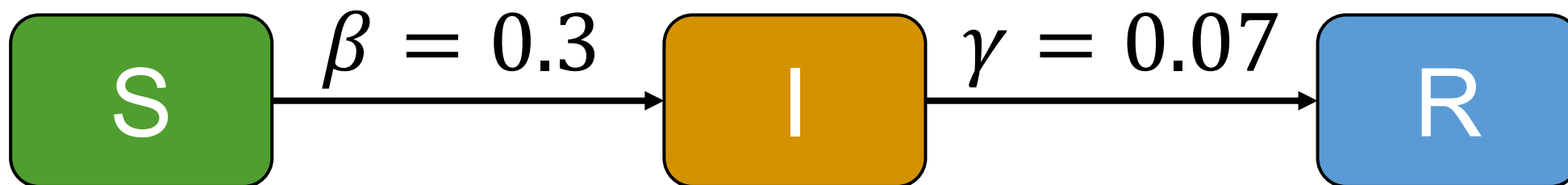


2. Introduction to individual-based models

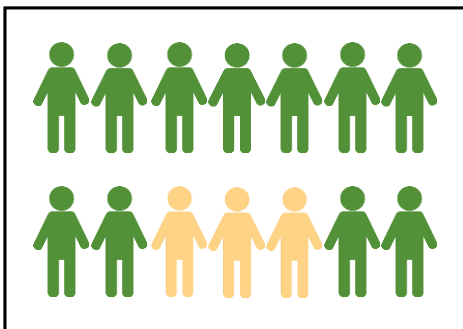


3. Introduction to spatial transmission models

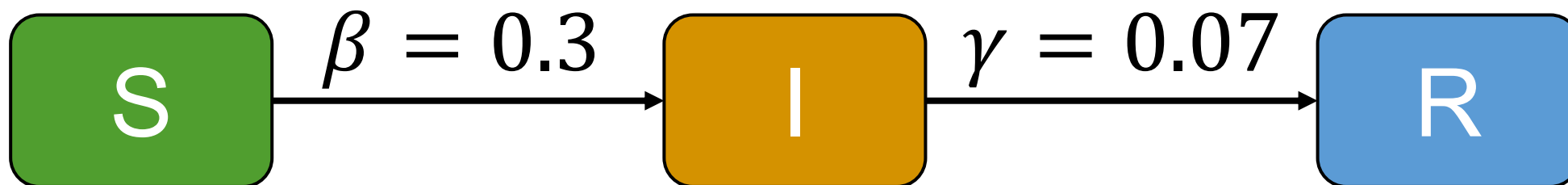
3. Introduction to spatial transmission models



*Non-spatial
transmission models*



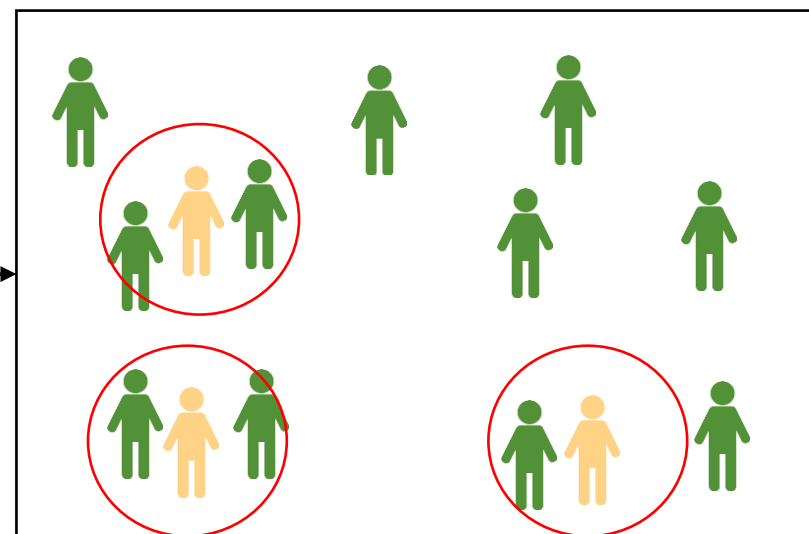
3. Introduction to spatial transmission models



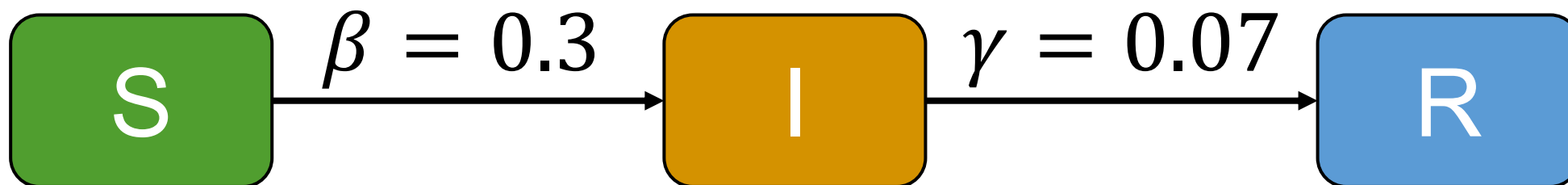
Non-spatial transmission models



Spatial transmission models



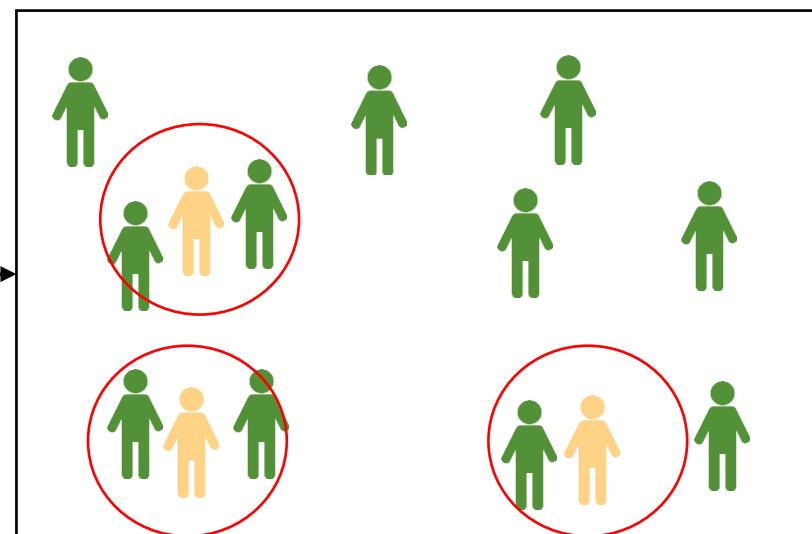
3. Introduction to spatial transmission models



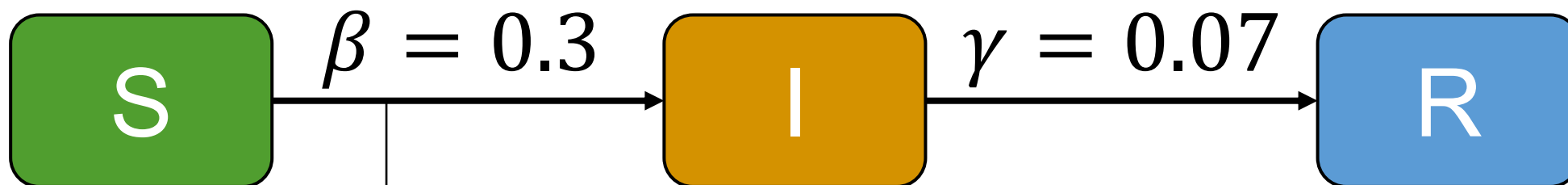
Non-spatial transmission models



Spatial transmission models



3. Introduction to spatial transmission models



Non-spatial
transmission models

$$P(\text{infection}) = 1 - (1 - \beta)^k$$

Spatial transmission
models

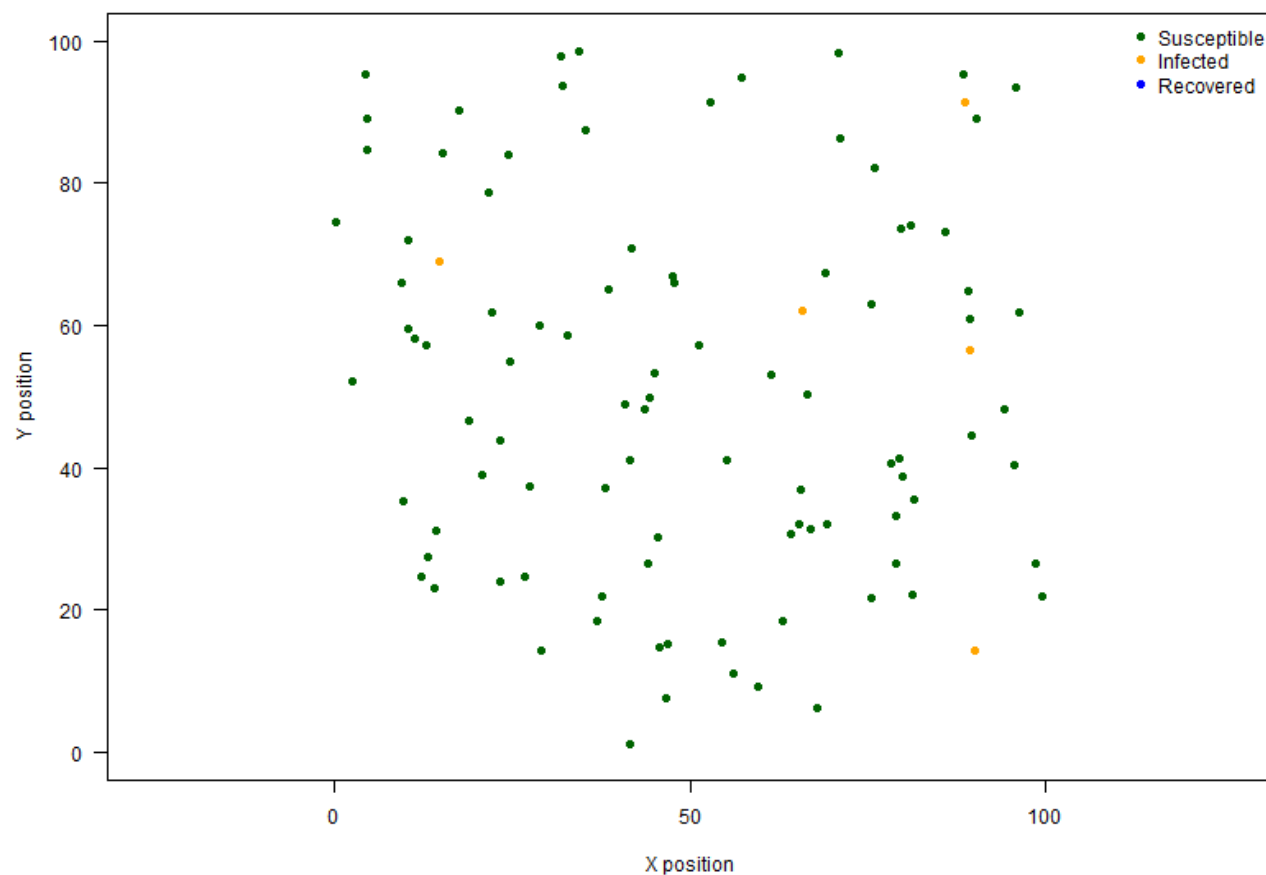
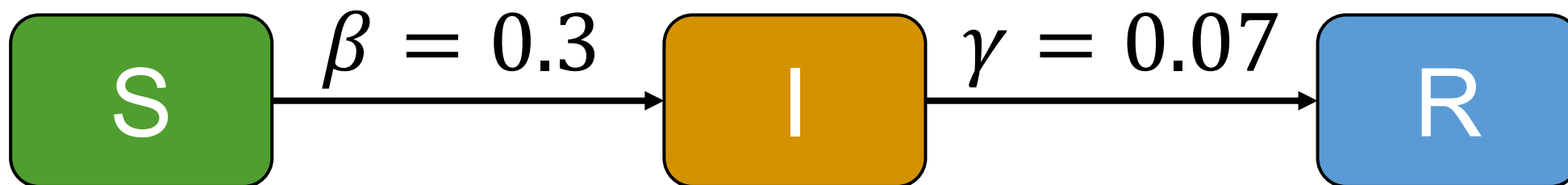
$$p_{ij} = \beta e^{-d_{ij}/e}$$

Distance

Spread

$$P(\text{infection}) = 1 - \prod_{j=1}^k (1 - p_{ij})$$

3. Introduction to spatial transmission models



3. *Introduction to spatial transmission models*

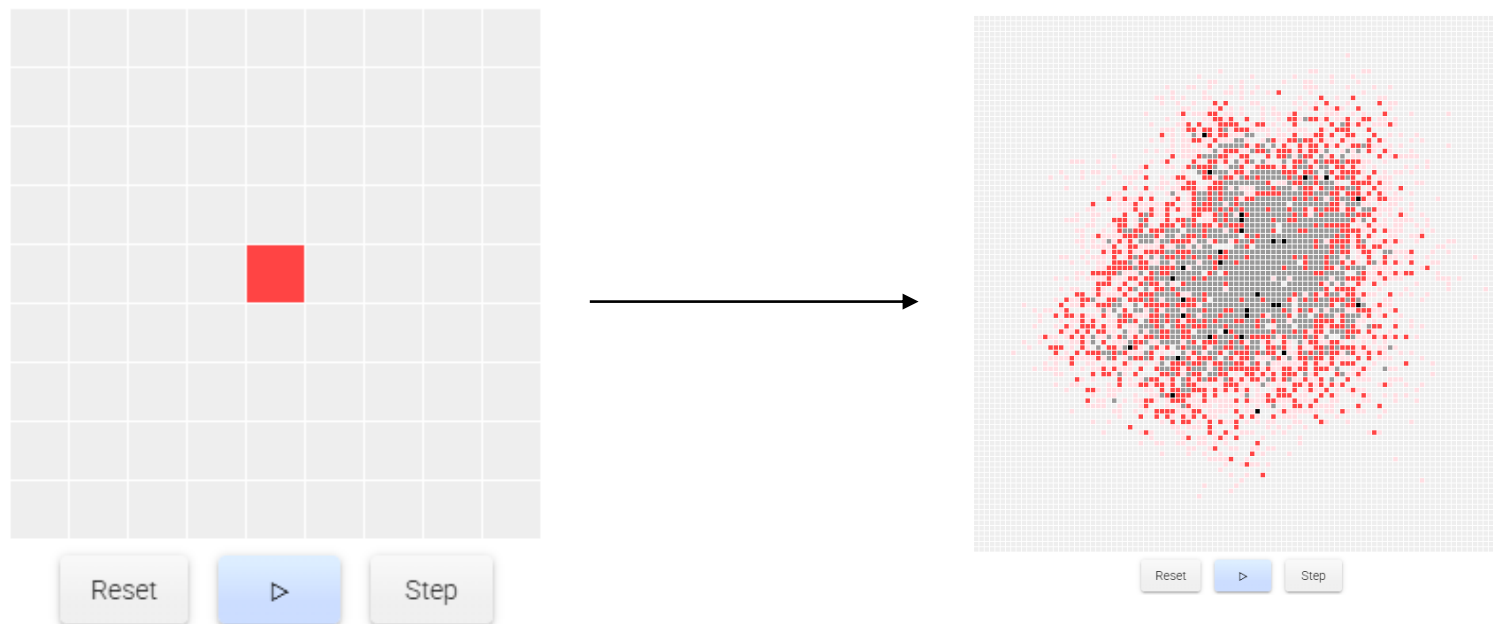
Time

Now let's incorporate time into our model.

The "Step" button (below) moves the simulation forward 1 day per click. Or you can press the ▷ button to watch things happen on their own:



Kevin Simler

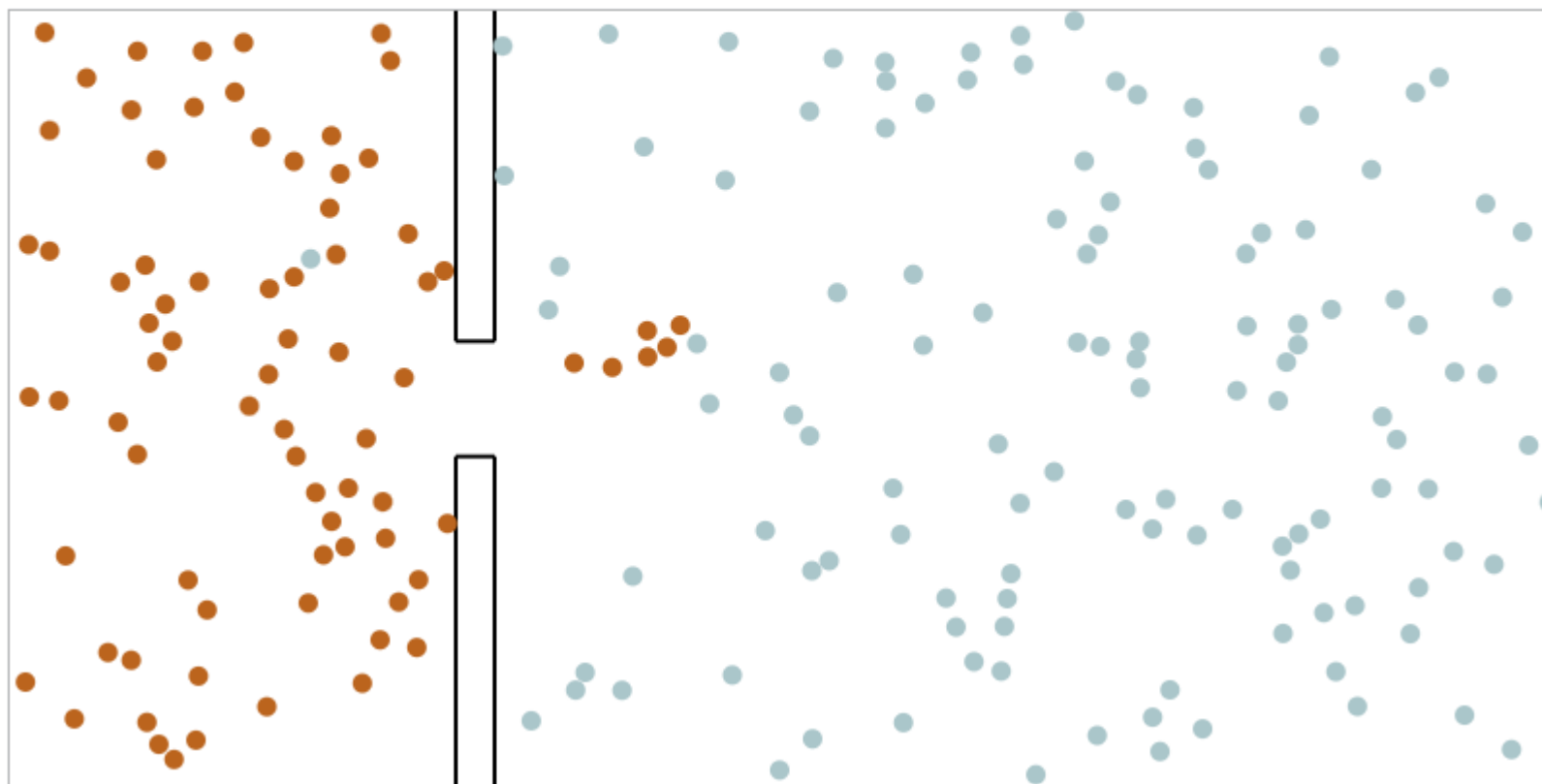
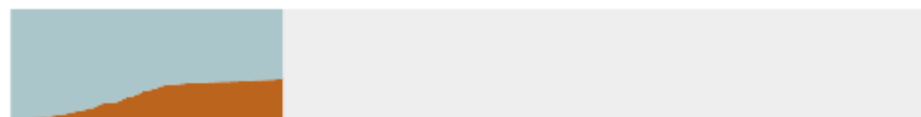


3. Introduction to spatial transmission models

Count

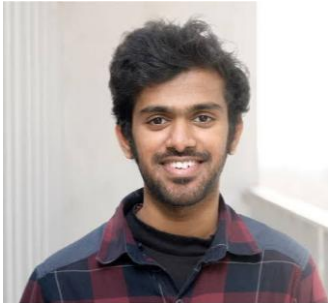
Recovered	0
Healthy	122
Sick	78

Change over time



Harry Stevens

3. *Introduction to spatial transmission models*

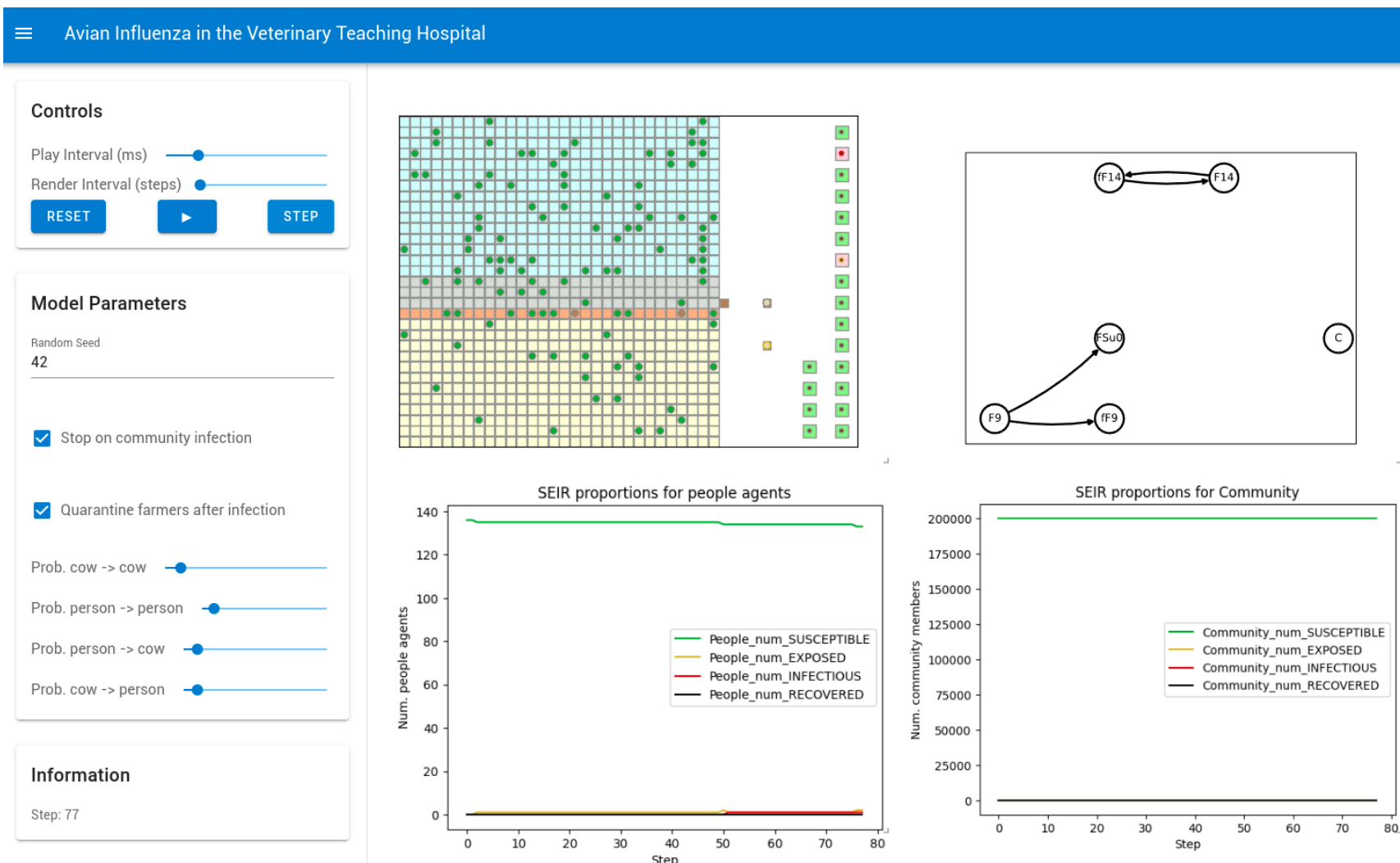


Prajwal Souza



3. Introduction to spatial transmission models

Model of Avian Influenza around a Veterinary Teaching Hospital

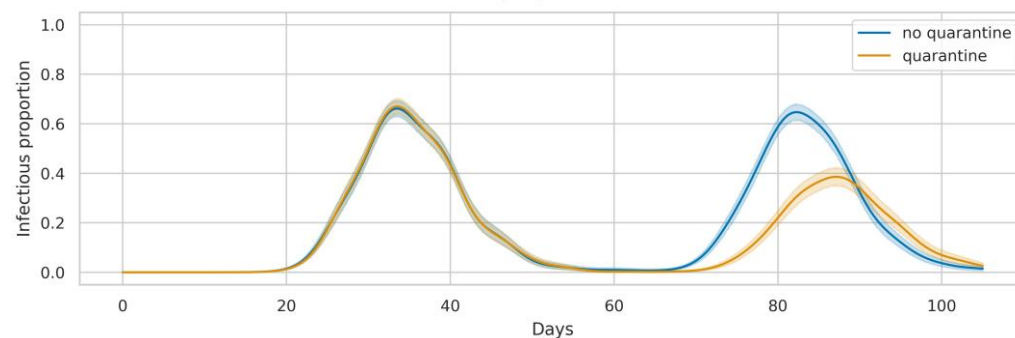


Gregor McEwan

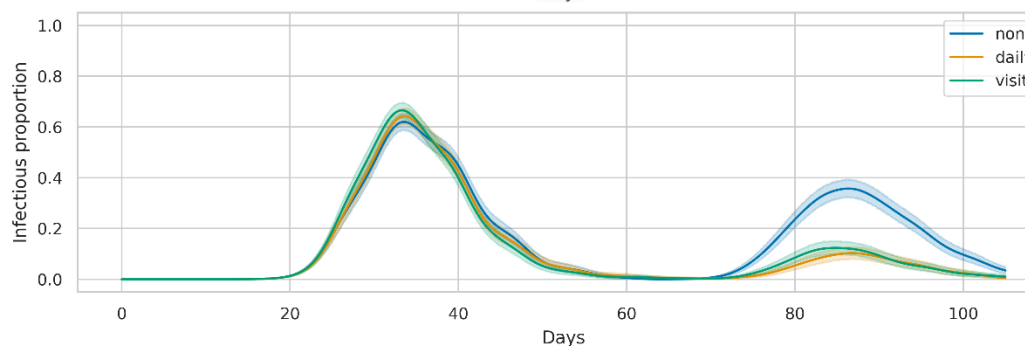
(McEwan et al., 2026)

3. Introduction to spatial transmission models

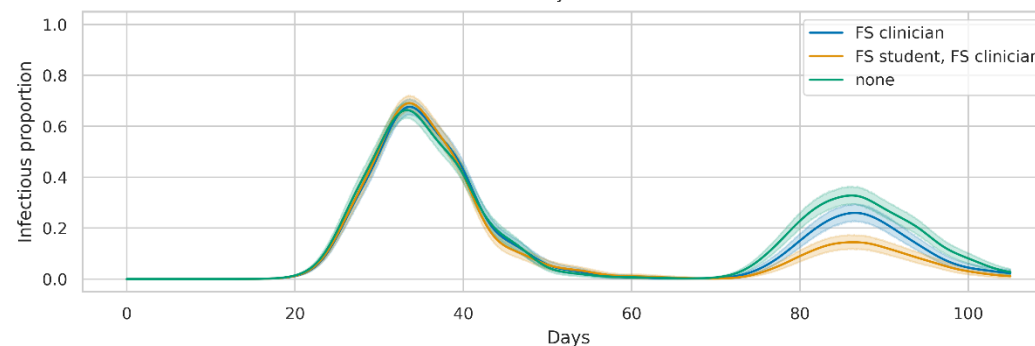
Model of Avian Influenza around a Veterinary Teaching Hospital



Quarantine farmers



Cleaning Trucks



Vaccination



Gregor McEwan

(McEwan et al., 2026)

3. *Introduction to spatial transmission models*



Matthew Silk



3. Introduction to spatial transmission models



Matthew Silk



3. Introduction to spatial transmission models

YOUR TIME AS A NERGAL



You're **less sociable** than most Nergals.
You talked to 2 Nergals, and made 2 friends.



You're **more sickly** than most Nergals.
You lived for 247 seconds, got sick 3 times and infected 2 other Nergals.



You talked to **fewer clouds** than most Nergals.



Playing the game generated important data for research on how diseases spread, the

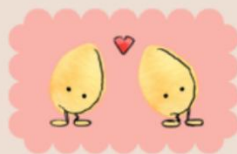
More about your data

Play again

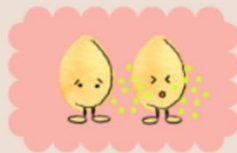


3. Introduction to spatial transmission models

YOUR TIME AS A NERGAL



You're **more sociable** than most Nergals.
You talked to 5 Nergals, and made 0 friends.



You're **more sickly** than most Nergals.
You lived for 250 seconds, got sick 4 times
and infected 5 other Nergals.



You talked to **fewer clouds** than most Nergals.



Playing the game generated important data for research on how diseases spread, thank you!

More about your data

Play again



3. *Introduction to spatial transmission models*

YOUR TIME AS A NERGAL



You're **more sociable** than most Nergals.
You talked to 20 Nergals, and made 0 friends.



You're **less sickly** than most Nergals.
You lived for 300 seconds, got sick 1 time and infected 2 other Nergals.



You were **less hungry** than most Nergals, eating 0 snacks.



Playing the game generated important data for research on how diseases spread

More about your data

Play again



- *Essential*

- Keeling, M.J., Rohani, P., 2011. Modeling infectious diseases in humans and animals, Modeling Infectious Diseases in Humans and Animals. [https://doi.org/10.1016/s1473-3099\(08\)70147-6](https://doi.org/10.1016/s1473-3099(08)70147-6)
- Vynnycky, E., White, R.G., 2010. An introduction to infectious disease modelling. Oxford University Press, New York. <https://doi.org/10.1093/eurpub/ckq163>

- *Further*

- McEwan, G., Stull, J., Marina, H., Sánchez, J. 2026. Model of Avian Influenza around a Veterinary Teaching Hospital. *Preprint*. <http://dx.doi.org/10.2139/ssrn.6431928>
- Kirkeby, C., Halasa, T., Gussmann, M., Toft, N., Græsbøll, K., 2017. Methods for estimating disease transmission rates: Evaluating the precision of Poisson regression and two novel methods. Scientific Reports. <https://doi.org/10.1038/s41598-017-09209-x>
- Woudstra, S., Wente, N., Zhang, Y., Leimbach, S., Gussmann, M.K., Kirkeby, C., Krömker, V., 2023. Strain diversity and infection durations of Staphylococcus spp. and Streptococcus spp. causing intramammary infections in dairy cows. Journal of Dairy Science. <https://doi.org/10.3168/JDS.2022-22942>
- McEwan, G., Stull, J., Marina, H., Sánchez, J. 2026. Model of Avian Influenza around a Veterinary Teaching Hospital. <http://dx.doi.org/10.2139/ssrn.6431928>

hands
On





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LIFE